The Delta Method: Uncertainty for Derived Quantities

What it is, where it comes from, and why it works (an NRES 746 reference handout)

A fitted model hands you estimates of its parameters and, from the Hessian, their variances and covariances. The number you care about is often something else: a prediction, a ratio, a back-transformed parameter, a population growth rate built from several vital rates. The **delta method** turns the uncertainty in the parameters into an approximate standard error for any smooth function of them. It needs two things: the variance-covariance matrix of the estimates, and the derivatives of the function. Bolker covers it in Section 7.5.2 and Appendix A.7 of *Ecological Models and Data in R*; this handout adds the derivation and a geometric picture of why it works.

The background assumed is derivatives (including partial derivatives) and the rules for the variance of a sum. The running example is the power-law fit to R's `trees` data, timber volume V against diameter D for 31 black cherry trees:

$$V_i \sim \text{Normal}\left(aD_i^b,\ \sigma^2\right).$$

Fitting by maximum likelihood with `optim()`, estimating $\log\sigma$ so that σ stays positive, gives $\hat{a} = 0.087$, $\hat{b} = 2.24$, $\widehat{\log\sigma} = 1.157$, and, from the inverse of the Hessian, standard errors 0.021, 0.086, and 0.127. The correlation between $\hat{a}$ and $\hat{b}$ is -0.998.

1. Why we need it

Two derived quantities come up with this model.

- **The residual SD.** We estimated $\log\sigma$, but we want to report $\sigma = e^{\log\sigma} = 3.18$ cubic feet, with a standard error. The Hessian only gives the standard error of $\log\sigma$.
- **A prediction.** The expected volume of a 13-inch tree is $\hat{a}\cdot 13^{\hat{b}} = 26.84$ cubic feet. How uncertain is it?

The prediction shows why a shortcut fails. Both $\hat{a}$ and $\hat{b}$ are uncertain, so the obvious move is to combine their two variances. Treating them as independent gives a standard error of 8.73 cubic feet (calculated in Section 2), a third of the prediction itself. That's absurd for a curve that runs through the middle of 31 tightly scattered points. The missing piece is the correlation of -0.998: when the data push $\hat{a}$ up, they push $\hat{b}$ down, and the two changes nearly cancel where the data are. The delta method keeps that covariance and gives 0.66 cubic feet.

Other places the same question comes up:

- back-transforming any parameter fit on a log, logit, or atanh scale;
- a ratio of curve parameters, such as the initial slope a/b of a Michaelis-Menten curve $ax/(b+x)$;
- the predictor value where a logistic curve crosses 0.5, which is $-\beta_0/\beta_1$;
- mean survival time from a Weibull fit (Bolker's example in 7.5.2);
- standard errors of GLM predictions on the response scale. `predict(..., type = "response", se.fit = TRUE)` uses the delta method behind the scenes.

2. The formula

Bolker's notation. For a function $f(a, b)$ of two estimates (Bolker's Eq. 7.5.4, with each derivative evaluated at the estimates):

$$\operatorname{Var}\Big(f(\hat{a},\hat{b})\Big) \approx \operatorname{Var}(\hat{a})\left(\frac{\partial f}{\partial a}\right)^2 + \operatorname{Var}(\hat{b})\left(\frac{\partial f}{\partial b}\right)^2 + 2\operatorname{Cov}(\hat{a},\hat{b})\,\frac{\partial f}{\partial a}\frac{\partial f}{\partial b}.$$

With one parameter, only the first term is left: $\operatorname{Var}(f(\hat{\theta})) \approx f'(\hat{\theta})^2 \operatorname{Var}(\hat{\theta})$, so $\operatorname{SE}(f) \approx |f'(\hat{\theta})|\operatorname{SE}(\hat{\theta})$.

Matrix notation. For k parameters, collect the partial derivatives into a column vector, the gradient $\mathbf{g} = (\partial f/\partial\theta_1, \ldots, \partial f/\partial\theta_k)^T$, and write Σ for the variance-covariance matrix of the estimates (the inverse of the Hessian, `solve(fit$hessian)` after `optim()`). Then

$$\operatorname{Var}\Big(f(\hat{\boldsymbol{\theta}})\Big) \approx \mathbf{g}^T \Sigma\, \mathbf{g} = \sum_{i=1}^{k}\sum_{j=1}^{k} g_i\, g_j\, \Sigma_{ij}.$$

The double sum has a squared derivative times a variance for each diagonal entry ($i = j$) and a cross term for each off-diagonal pair. With $k = 2$, the two off-diagonal entries are equal and give Bolker's $2\operatorname{Cov}$ term, so the two notations say the same thing.

One parameter: the residual SD. With $f(\theta) = e^\theta$ and $\theta = \log\sigma$, the derivative is $e^\theta = \sigma$, so

$$\operatorname{SE}(\hat{\sigma}) \approx \hat{\sigma} \cdot \operatorname{SE}(\widehat{\log\sigma}) = 3.18 \times 0.127 = 0.404.$$

Fitting σ directly and inverting that Hessian gives the same number. At the MLE, the two curvatures differ by exactly the squared derivative, so estimating on the log scale costs nothing.

Two parameters: the prediction at 13 inches. For $f(a, b) = a \cdot 13^b$,

$$\frac{\partial f}{\partial a} = 13^b = 309.9, \qquad \frac{\partial f}{\partial b} = a\,13^b \log 13 = 68.8.$$

Then

$$\mathbf{g}^T\Sigma\,\mathbf{g} = 309.9^2(4.29\times10^{-4}) + 68.8^2(0.00739) + 2(309.9)(68.8)(-0.00178) = 41.216 + 35.036 - 75.814 = 0.438,$$

a standard error of 0.66. The two variance terms are large; the negative covariance term cancels nearly all of them. Treating $\hat{a}$ and $\hat{b}$ as independent means dropping that term:

$$41.216 + 35.036 = 76.252, \qquad \sqrt{76.252} = 8.73,$$

the inflated standard error from Section 1.

In R, `emdbook::deltavar()` does this calculation from a formula, and `numericDeriv()` or a finite difference can stand in for derivatives you'd rather not work out by hand.

3. Where it comes from

The delta method is a first-order Taylor expansion followed by the variance of a linear function.

One parameter. Near the true value θ_0, replace f with its tangent line:

$$f(\hat{\theta}) \approx f(\theta_0) + f'(\theta_0)\,(\hat{\theta} - \theta_0).$$

The right side is a constant plus a constant times $\hat{\theta}$. Adding a constant doesn't change a variance, and multiplying by c multiplies it by c^2, so

$$\mathrm{Var}\Big(f(\hat{\theta})\Big) \approx f'(\theta_0)^2\,\mathrm{Var}(\hat{\theta}).$$

We don't know θ_0, so the derivative is evaluated at $\hat{\theta}$, which is close to it when the standard error is small.

Several parameters. The tangent line becomes a tangent plane:

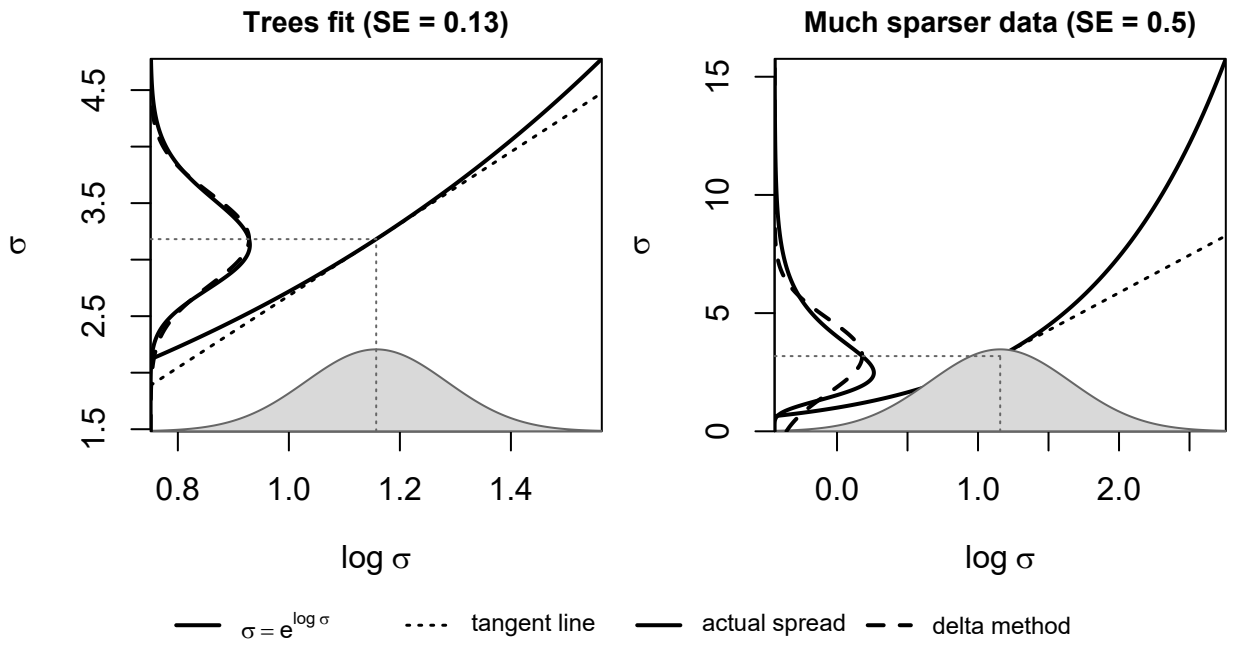
$$f(\hat{\boldsymbol{\theta}}) \approx f(\boldsymbol{\theta}_0) + \sum_i g_i\,(\hat{\theta}_i - \theta_{0,i}) = f(\boldsymbol{\theta}_0) + \mathbf{g}^T(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0).$$

The variance of a weighted sum $\sum_i g_i X_i$ is $\sum_i \sum_j g_i g_j\,\mathrm{Cov}(X_i, X_j)$, which is $\mathbf{g}^T \Sigma\, \mathbf{g}$. For two parameters, $\mathrm{Var}(g_a X_a + g_b X_b) = g_a^2 \mathrm{Var}(X_a) + g_b^2 \mathrm{Var}(X_b) + 2 g_a g_b \mathrm{Cov}(X_a, X_b)$, which is Bolker's formula.

The shape of the distribution, too. The MLEs are approximately multivariate normal (see the companion handout below), and a linear function of normal variables is normal. So to the same approximation, $f(\hat{\boldsymbol{\theta}}) \approx \mathrm{Normal}\Big(f(\boldsymbol{\theta}_0),\ \mathbf{g}^T \Sigma\, \mathbf{g}\Big)$, and a 95% interval is $f(\hat{\boldsymbol{\theta}}) \pm 1.96\,\mathrm{SE}$. For the 13-inch tree, that's [25.5, 28.1] cubic feet.

Everything rests on one assumption: **over the range where the estimates plausibly fall, f is close to its tangent line (or plane).** The smaller the standard errors, the smaller that range, and the better the approximation.

4. The geometry with one parameter



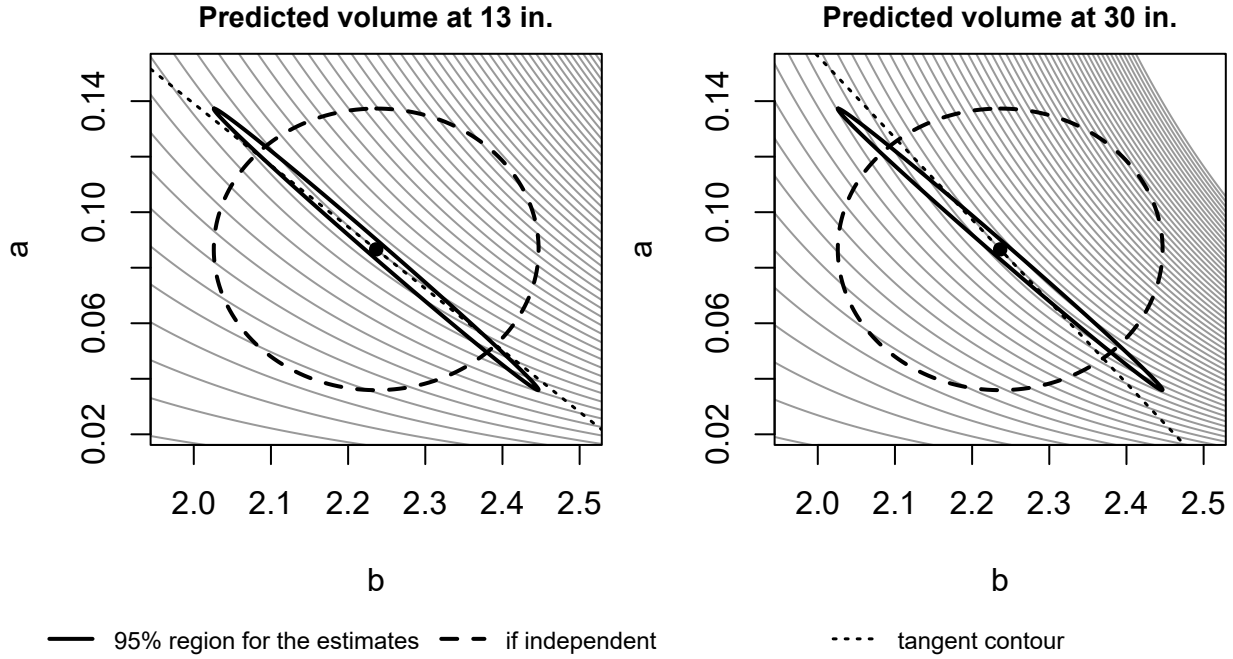
Read each panel from the bottom up.

- **Bottom (grey):** the sampling distribution of $\widehat{\log \sigma}$, normal with the standard error from the Hessian.
- **The curve** carries each possible estimate up to a value of σ. The spread of the estimates along the bottom becomes a spread of $\hat{\sigma}$ along the side. The solid curve on the left edge is that actual spread.
- **The tangent line** at the estimate is the delta method's stand-in for the curve. A line stretches every distance by the same factor, its slope, so it turns a normal distribution into a normal distribution with an SD multiplied by $|f'|$. The dashed curve on the left edge is that result.

On the left, the estimates fall in a narrow band where the curve and its tangent are hard to tell apart, and the two side curves nearly coincide. On the right, the band is wide enough that the curve bends away from its tangent. The upper half of the band is stretched more than the lower half, so the actual spread is skewed to the right, and the symmetric delta-method normal misses it. The slope of f sets the width of the result; the bending of f over the band sets how wrong the delta method is.

5. The geometry with two parameters

With two parameters, the sampling distribution is a tilted ellipse, and the function is a landscape over the (b, a) plane. Its contour lines connect parameter pairs that give the same prediction.



- **The solid ellipse** is where the estimates $(\hat{b}, \hat{a})$ plausibly fall (a 95% region from Σ). It is long and thin and runs downhill: the data constrain the combination of a and b far better than either alone.
- **The grey contours** are levels of the prediction, every 2 cubic feet on the left and every 10 on the right. Within a panel, the uncertainty in the prediction is how many contours the ellipse crosses.
- **Left, 13 inches (the middle of the data).** The ellipse lies almost along a contour, so it crosses only one or two: SE 0.66 cubic feet. The dashed ellipse is what you'd assume if $\hat{a}$ and $\hat{b}$ were independent. It cuts straight across the contours, which is the inflated SE of 8.73 from Section 1. The covariance term is what tilts the ellipse.
- **Right, 30 inches (beyond the largest tree, 20.6).** The contours have rotated: the dotted tangent contour now crosses the ellipse at an angle instead of running along it, so moving along the ellipse changes the prediction. SE 9.82 cubic feet, about 6% of the prediction instead of 2.5%. Extrapolating points the prediction in a direction the data don't constrain.

What the delta method does in this picture. It replaces the curved contours with straight, evenly spaced lines parallel to the tangent contour (dotted). The gradient $\mathbf{g}$ points straight across these lines, and its length says how close together they are. $\mathbf{g}^T \Sigma \, \mathbf{g}$ is then the variance of the ellipse measured in that one direction: squash the ellipse onto the gradient's direction, take the variance of that shadow, and scale by $|\mathbf{g}|^2$. That's why the cross term matters. An ellipse's shadow depends on its tilt, and the tilt is the covariance.

6. Checking it, and when it breaks

A simulation check. Simulate 2,000 new data sets from the fitted model, refit each one, and compute the prediction from each refit (a parametric bootstrap). The SD of those predictions is a check that doesn't use the delta method:

Diameter (in)	Prediction (cubic ft)	Delta-method SE	SE ignoring covariance	Bootstrap SD
8	9.1	0.56	2.71	0.57
13	26.8	0.66	8.73	0.67
21	78.4	2.24	27.82	2.26
30	174.2	9.82	65.80	9.87

The delta method matches the bootstrap closely across the data and at 30 inches. Leaving out the covariance overstates the uncertainty by a factor of 5 to 13.

When it breaks. Each case is a way for f to stop looking like its tangent over the plausible range of the estimates.

- **Large standard errors relative to the curvature of f.** That's the right panel of the Section 4 figure: the real distribution of $f(\hat{\theta})$ is skewed, and ± 1.96 SE is symmetric. Bolker's Weibull example (7.5.5) shows the cost: for mean survival time, the delta-method interval is [7.4, 22.4] and the profile likelihood interval is [9.2, 27.0].
- **A derivative near zero.** At a peak or trough of f, the tangent is flat and the delta method reports almost no uncertainty, whatever the estimates do.
- **Ratios with an uncertain denominator.** For $-\beta_0/\beta_1$, if the interval for β_1 comes near 0, the ratio can be huge or change sign, and no straight line describes it.
- **Bad covariances.** The method is only as good as Σ. If the Hessian approximation is poor (small samples, boundaries, the wrong model), the delta method inherits the problem.

What to do instead.

- **Change the scale first.** For a positive quantity, apply the delta method to $\log f$, build the interval there, and exponentiate the endpoints. The result stays positive and is skewed in the right direction. For σ: $\exp(1.157 \pm 1.96 \times 0.127)$ gives [2.48, 4.08], compared with [2.39, 3.97] on the raw scale.
- **Profile likelihood** for f, by reparameterizing so f is a parameter (Bolker 7.5.1).
- **Simulate.** Use a parametric bootstrap, as above, or draw parameter sets from $\text{MVN}(\hat{\boldsymbol{\theta}}, \Sigma)$ and compute f for each (Bolker's population prediction intervals, 7.5.3). The second option inherits the normal approximation's shape for the parameters, so its answer depends on the scale they're on. Here, drawing (a, b) from the MVN gives an SD of 1.51 cubic feet at 13 inches, against the bootstrap's 0.67. The real sampling distribution of $(\hat{b}, \hat{a})$ bends along the curved contours in the figure, and a straight ellipse can't follow it. On the $(\log a, b)$ scale, where those contours are straight lines, the draws give 0.66.

- **Go Bayesian.** With posterior draws, compute f for each draw and summarize; nothing needs to be linearized.

A note on Bolker's Appendix A.7. It lists the product of two independent variables as a case where the formula is exact. It isn't quite: the exact variance has an extra term, $\mathrm{Var}(XY) = \mu_Y^2 \mathrm{Var}(X) + \mu_X^2 \mathrm{Var}(Y) + \mathrm{Var}(X)\mathrm{Var}(Y)$. The delta method drops the last term, which is small when both coefficients of variation are small. The formula is exact for linear functions, as A.7 also says.

Summary

Idea	Picture	Formula
One parameter	tangent line stretches the spread by its slope	$\mathrm{SE}(f) \approx \lvert f'(\hat{\theta})\rvert \, \mathrm{SE}(\hat{\theta})$
Two parameters (Bolker)	spread of the ellipse across the contours	$\mathrm{Var}(\hat{a}) f_a^2 + \mathrm{Var}(\hat{b}) f_b^2 + 2\,\mathrm{Cov}(\hat{a}, \hat{b}) f_a f_b$, with $f_a = \partial f / \partial a$
k parameters	shadow of the ellipse on the gradient direction	$\mathbf{g}^T \Sigma \, \mathbf{g}$
Where it comes from	first-order Taylor expansion	$f(\hat{\theta}) \approx f(\theta_0) + \mathbf{g}^T (\hat{\theta} - \theta_0)$
Interval	normal, from the normal MLE	$f(\hat{\theta}) \pm 1.96\,\mathrm{SE}$
When it fails	f bends over the plausible range	use a log scale, profile, or simulate

The delta method assumes the function is straight over the range where the estimates fall, the same kind of assumption as the parabola behind Hessian intervals. A companion handout, *Why Likelihood Intervals Work*, explains where Σ comes from and why the MLE is approximately normal.

References

Bolker, B. M. 2008. *Ecological Models and Data in R.* Princeton University Press, Princeton, NJ. (Section 7.5 and Appendix A.7.)

Oehlert, G. W. 1992. A note on the delta method. *The American Statistician* 46: 27-29.

Ver Hoef, J. M. 2012. Who invented the delta method? *The American Statistician* 66: 124-127.